

# Joonsuk Bae

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*ALICE PWG-JE Run 3 charged-particle jets — end-to-end pipeline for the first inclusive cross-section measurement at  $\sqrt{s} = 13.6$  TeV (preliminary; paper in preparation; EPS-HEP 2025 PWG-JE merge talk on behalf of ALICE).  
Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.*

## Education

**Ph.D. Candidate, Physics** — Sungkyunkwan University 2022.03 – expected 2027.02

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE*. Defense expected 2026.11–12; degree conferral 2027.02.

**B.S., Physics** — Sungkyunkwan University 2021.02

## Appointments and Collaboration Membership

|                                                    |                   |
|----------------------------------------------------|-------------------|
| ALICE Collaboration, CERN, Geneva                  | 2022.10 – present |
| ePIC Collaboration, Brookhaven National Laboratory | 2024.02 – present |
| LAMPS Collaboration, RAON, Korea                   | 2022.03 – 2024.04 |

## Awards and Fellowships

- **BK21 FOUR Graduate Scholarship** — Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.
- **Conference travel support** — KoALICE, NRF Korea–CERN program, and SKKU Graduate School; HP 2024 (Nagasaki), QM 2025 (Frankfurt), EPS-HEP 2025 (Marseille).

## Research Experience

**Main analyzer, ALICE Run 3 charged-particle jets (PWG-JE)** — CERN 2023.04 – present

- End-to-end ownership of the first ALICE Run 3 inclusive charged-particle jet cross-section measurement at  $\sqrt{s} = 13.6$  TeV — O<sup>2</sup>Physics task development, QA, Bayesian/SVD unfolding, full systematic-uncertainty program, and PWG-JE-approved preliminary results. Paper in preparation; presented on behalf of ALICE at EPS-HEP 2025.
- **Tracking-efficiency systematics for Hard Probes 2026 (PWG-JE)** — contributed the pp  $\sqrt{s} = 13.6$  TeV charged-particle tracking-efficiency systematic ( $p_T$ -binned,  $\sim 1.5\%$  across the dominant range) for the 2022, 2023, and 2024 periods; the ITS-TPC matching code was reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **Trigger-level inefficiency in the new readout** — quantified bunch-crossing and track-matching inefficiencies introduced by the continuous-readout TPC and asynchronous reconstruction (each with its own time-frame); derived signal-loss corrections used in subsequent PWG-JE Run 3 charged-jet analyses.
- **Track  $p_T$  resolution** — studied data–MC resolution mismatch via covariance-matrix propagation, MC residuals, and V0 mass-resolution methods; tuned the central track momentum-resolution smearing methodology with a  $p_T$ -dependent profile used in Run 3 charged-jet systematic studies.
- **Cross-section normalization** — established the Run 3 charged-jet normalization based on the TVX visible cross section under continuous readout; authored the PWG-JE cross-section normalization framework (PR #14057, merged) used at particle level by the SKKU Run 3 charged-jet analyses (pp 13.6 TeV and O+O). Provided the normalization factor and QA for the ALICE Run 3 O+O charged-jet measurement led by two MSc co-analyzers in the SKKU group.
- Presented preliminary results at **EPS-HEP 2025** (on behalf of the ALICE Collaboration; PWG-JE merge talk), **Quark Matter 2025**, and **Hard Probes 2024**; regular PWG-JE working-group speaker since 2023.
- Extending the program to semi-inclusive h+jet recoil (PWG-JE Run 3),  $R$ -dependent cross-section ratios as a hadronization-vs-perturbative-radiation probe, and groomed substructure (Soft Drop  $z_g$ ; primary Lund-plane

density) in high-multiplicity pp and forthcoming O+O / Ne+Ne running, motivated as a small-system baseline for energy-loss searches.

**Detector physicist, ePIC Barrel Imaging Calorimeter (BIC)** — Korea-BIC consortium 2024.02 – present

- Led PMT-module QA prior to test beams — cosmic-ray and source signal characterization, timing-coincidence checks, optical-coupling stability; applied per-module acceptance criteria.
- On-site DAQ shifter at **KEK AF-AR** (2025.03; equalization, prompt analysis; co-author on the corresponding paper in preparation) and lead offline analyst for the **CERN PS T10** (2025.07) campaign of the Pb/scintillating-fiber prototype — energy response, resolution, and linearity (in preparation, 2026).
- Contributed energy- and  $\eta$ -dependent sampling-fraction studies in JANA2 / EIC-recon for the Korea-BIC simulation effort; Geant4 prototype modeling.
- Module assembly and integration — Pb/scintillating-fiber bundle alignment, polishing, PMT/SiPM coupling.

**Detector R&D, LAMPS Forward Tracker** — RAON, Korea 2022.03 – 2024.04

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

## Selected Publications

Full publication list: [inspirehep.net/authors/2117232](https://inspirehep.net/authors/2117232). ALICE Collaboration co-author since 2022.10.

### First/lead-analyst works in preparation

- **FLAGSHIP** ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* (in preparation, target 2026). *First ALICE Run 3 inclusive charged-jet publication; thesis paper.* [main analyzer; full responsibility for analysis, systematics, unfolding, and paper draft]
- Korea-BIC Collaboration, *Beam-test characterization of a Pb/scintillating-fiber imaging calorimeter prototype — CERN PS T10 (2025.07) campaign* (in preparation, 2026). [lead offline analyst; on-site prompt analysis and follow-up offline reconstruction]

### Submitted (detector)

- Y. Hong, J. Bok, G. An, **J. Bae et al.**, “Test-beam performance of the AstroPix silicon sensor for imaging calorimetry,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). Corresponding: Prof. Sanghoon Lim. [co-author; ePIC BIC silicon-layer test-beam analysis]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae et al.**, “Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider” — first beam-test campaign; submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; also submitted to JKPS); arXiv preprint posted. Corresponding: H. Lee. [co-author; module-construction support and PMT/module performance characterization (KPS Spring 2025 poster)]

### In preparation (detector — co-author roles)

- Korea-BIC Collaboration, *Beam-test characterization of a Pb/scintillating-fiber imaging calorimeter prototype — KEK AF-AR (2025.03) campaign* (in preparation, 2026). [co-author; DAQ shifter and on-site prompt-analysis at KEK test beam]

### ePIC/EIC and related calorimeter R&D (peer-reviewed)

- J. K. Adkins et al. (ECCE Collaboration), “Design of the ECCE detector for the Electron-Ion Collider,” *Nucl. Instrum. Meth. A* **1073** (2025) 170240. [co-author; ECCE consortium member]
- S. Y. Jang, G. An, **J. Bae et al.**, “Performance of dual-readout calorimeters for various absorbers,” *Nucl. Instrum. Meth. A* (2026), DOI 10.1016/j.nima.2026.171598. [co-author; dual-readout R&D for  $e^+e^-$  colliders]

- G. Cho, S. Kim, **J. Bae et al.**, “Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future  $e^+e^-$  colliders,” *J. Subatom. Part. Cosmol.* **3** (2025) 100021. [co-author; module construction and beam-test data taking]
- ECCE Consortium, “AI-assisted optimization of the ECCE tracking system at the Electron-Ion Collider,” *Nucl. Instrum. Meth. A* (2023), DOI 10.1016/j.nima.2022.167748. [co-author; ECCE consortium member]
- ECCE Consortium, “Open heavy-flavor studies for the ECCE detector at the Electron-Ion Collider,” arXiv:2207.10632 (2022). [co-author; preprint]

### Earlier detector R&D (LAMPS, RAON; peer-reviewed)

- Y. J. Kim, D. S. Ahn, J. K. Ahn *et al.* (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682. [co-author; Time-of-Flight detector maintenance and on-site checks]
- Y. Bae *et al.* (LAMPS), “Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV,” *Nucl. Instrum. Meth. A* (2025), DOI 10.1016/j.nima.2025.170827. [co-author; Start-Counter prototype testing]
- Y. Cheon *et al.* (LAMPS), “Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON,” *Nucl. Instrum. Meth. A* **1066** (2024) 169610. [co-author; TPC prototype characterization]
- H. Kim *et al.* (LAMPS), “Performance of the prototype beam drift chamber for LAMPS at RAON with proton and  $^{12}\text{C}$  beams,” *JINST* **19** (2024) P12008; arXiv:2412.08662. [co-author; forward-tracking R&D and prototype performance studies]
- LAMPS Collaboration, “Status of large acceptance multi-purpose spectrometer (LAMPS) at RAON,” *AIP Conf. Proc.* **2319** (2021) 080002. [co-author; SiPM and scintillator-based detector R&D]

## Software and Computing

### ALICE O<sup>2</sup>Physics — AliceO2Group/O2Physics

- **Central-framework fix** to the track momentum-resolution smearing methodology shared across analyses (PR #15174, merged).
- **Authored the PWG-JE cross-section normalization framework** (PR #14057, merged) used by the SKKU Run 3 charged-jet analyses for pp 13.6 TeV and O+O.
- 37 pull requests, 18 merged (2023 – present), covering jet finding, response matrices, systematic-uncertainty pipelines, hadronic-rate fixes, and central reconstruction.

## Invited and Contributed Talks

**Summary since 2022:** 3 invited workshops [external] · 5 invited collaboration-meeting talks [internal] · 6 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025) · 8 domestic (KPS) · regular PWG-JE working-group speaker since 2023.

### Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

### Invited Collaboration-Meeting Talks (internal)

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01

- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet  $O^2$  MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

### Contributed Talks and Posters

- **EPS-HEP 2025**, Marseille — *Inclusive and semi-inclusive jet cross-section measurements in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE [on behalf of the ALICE Collaboration; PWG-JE merge talk]* — 2025.07
- **INPC 2025**, Daejeon — *Charged-particle jet and jet quenching measurement in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV during LHC Run 3 with ALICE [international, talk]* — 2025.05
- **Quark Matter 2025**, Frankfurt — *Inclusive charged-particle jet spectrum in  $pp$  collisions in Run 3 at  $\sqrt{s} = 13.6$  TeV with ALICE [international, poster]* — 2025.04
- **Hard Probes 2024**, Nagasaki — *First measurements of charged-particle jet production in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE [international, poster]* — 2024.09
- **Quark Matter 2023**, Houston — *Charged-particle multiplicity distribution in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE [international, poster]* — 2023.09
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon — *Dijet studies at the LHC [international, poster]* — 2023.04
- **KPS Fall 2025**, Daejeon [domestic, talk] — *Inclusive and semi-inclusive jet cross-section measurements in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2025.10
- **KPS Spring 2025** [domestic, talk] — *Inclusive charged-particle jet spectrum in  $pp$  collisions in Run 3 at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2025.04
- **KPS Spring 2025** [domestic, poster] — *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* — 2025.04
- **KPS Fall 2024** [domestic, talk] — *First jet measurement in ALICE Run 3: charged-particle jet production* — 2024.10
- **KPS Spring 2024** [domestic, talk] — *Charged-particle jet production in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV* — 2024.04
- **KPS Fall 2023** [domestic, talk] — *First look at charged-particle jet production in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE Run 3 data* — 2023.10
- **KPS Spring 2023** [domestic, poster] — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2022** [domestic, poster] — *Dijet studies with ALICE at the LHC* — 2022.10

## Test-beam and Detector Activities

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### ALICE ITS2 alignment monitoring at CTF level (Run 3)

- Analyzed compressed-time-frame (CTF) data — below the standard AO2D analyzer interface — to provide period-by-period tracking-performance diagnostics.

### ePIC Barrel Imaging Calorimeter test beams

- **CERN PS T10** (2025.07) — on-site analysis lead; energy response and resolution.
- **KEK AF-AR** (2025.03) — DAQ, equalization, prompt analysis.
- **Korea-BIC module production** (2024 – 2025) — Pb/scintillating-fiber bundle assembly, polishing, and PMT/SiPM integration; PMT-level QA on the prototype modules used in the 2025 beam-test campaigns.

## Service and Professional Activities

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- **PWG-JE tracking-efficiency contribution for Hard Probes 2026** —  $pp$  13.6 TeV studies; ITS-TPC matching code reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **ALICE Run 3 O+O charged-jet measurement** (PWG-JE) — co-author on the Analysis Note; service contribution: cross-section normalization factor derivation and QA; co-mentor of the two MSc analyzers.
- **ALICE ITS2 alignment monitoring** — service work at CTF level for tracking-performance diagnostics.

- **ePIC Korea-BIC analysis coordination** — analysis liaison between the SKKU group and the Korea-BIC consortium for beam-test data products.
- **KoALICE national activities** — regular participation since 2022, including invited collaboration-meeting talks.

## Teaching and Mentoring

**Student mentoring (non-supervisory)** — Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc., co-analyzer on the ALICE Run 3 O+O charged-jet measurement) and *one further MSc co-mentee* (co-analyzer on the same O+O analysis) — mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author with W. Ham on the corresponding PWG-JE Analysis Note.
- *Harim Seong* (undergraduate) — ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

**KoALICE O<sup>2</sup> Tutorial — student co-organizer and instructor** — Sungkyunkwan University, Suwon 2024.01.22 – 23

- Co-organized (with Hyungjun Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community — ~22 participants from SKKU, Inha University, and other Korean institutes; listed as the official Indico contact for the workshop. [indico.cern.ch/event/1361235](https://indico.cern.ch/event/1361235)
- Instructor on *Running O<sup>2</sup> Physics with Hyperloop* (solo) and co-instructor on *Practical PWG-JE tasks* (with H. Lee).

**Teaching Assistant**, SKKU Physics (2022 – 2024) — *Nuclear Reactions* (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I*.

## Technical Skills

**Languages.** English (professional working proficiency), Korean (native).

**Programming.** C++, Python, ROOT, L<sup>A</sup>T<sub>E</sub>X, Bash, Git.

**HEP frameworks.** ALICE O<sup>2</sup>Physics, Geant4, JANA2 / EIC-recon, PYTHIA, FastJet.

**Analysis methods.** jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.

**Detector hardware.** PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

## References

Available on request.